

TW03: Python Data Structures

Five challenges | 20 marks each | 100 marks total

Student name	
Student ID	
Class / group	
Submission date	

Weekly purpose

Organise and retrieve data with lists, tuples, dictionaries, sets, and nested records.

SUBMISSION RULES

- Complete work in the corresponding Jupyter notebook.
- Write your own code and explanations; no solutions are supplied.
- Keep supplied data unchanged and work on copies.
- Restart the kernel and run all cells before submission.
- Submit the notebook with outputs visible and assistance acknowledged.

MARKING RUBRIC FOR EACH CHALLENGE

Assessment area	Marks
Correct method and working code	8
Validation and quality checks	4
Explanation and interpretation	4
Reproducibility and presentation	4
Total per challenge	20

Challenge 1: Explore a weekly sales list

Scenario

A manager needs selected values and small changes from an ordered list of daily sales.

INSTRUCTIONS

1. Display the first, last, and third value using indexes.
2. Display the first three and final three values using slices.
3. Replace one incorrect value and append a new day.
4. Report length, total, minimum, maximum, and average.
5. Check whether the value 600 appears in the list.

REQUIRED SUBMISSION

- Correct indexed and sliced outputs.
- An updated list with the original preserved in a separate variable.
- Five labelled summaries and one membership result.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 2: Manage an inventory without losing the original

Scenario

A stock list must be cleaned, sorted, and expanded while keeping an untouched copy for audit.

INSTRUCTIONS

1. Create a true copy of the supplied list.
2. Add monitor, remove mouse, and replace keyboard with keyboard.
3. Create ascending and descending sorted versions without changing the working order.
4. Count how many times laptop occurs and report its first position.
5. Prove that the original list remains unchanged.

REQUIRED SUBMISSION

- Original, working, ascending, and descending lists.
- A laptop count and index.
- A Boolean audit check showing the original is preserved.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 3: Use tuples for fixed records

Scenario

A product code, name, and price form a small record that should not be changed accidentally.

INSTRUCTIONS

1. Create a tuple containing product code P104, name Webcam, and price 39.99.
2. Unpack the tuple into three clearly named variables.
3. Create a second tuple containing latitude and longitude values.
4. Use indexing to display only the longitude.
5. Explain one situation where a tuple is preferable to a list.

REQUIRED SUBMISSION

- Two valid tuples.
- Correctly unpacked product fields and indexed longitude.
- A justified list-versus-tuple choice.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 4: Maintain a customer dictionary

Scenario

A support team stores one customer's basic details as key-value pairs.

INSTRUCTIONS

1. Create keys for customer_id, name, city, membership, and orders.
2. Retrieve the name using brackets and retrieve phone safely with .get().
3. Update membership, add email, and remove city while saving the removed value.
4. Display keys, values, and items.
5. Check that customer_id is present and phone is absent.

REQUIRED SUBMISSION

- A correctly updated dictionary.
- Safe retrieval and saved removed value.
- Two membership checks and readable output.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 5: Combine sets and nested records

Scenario

The marketing team wants unique interests and a safely organised record for two customers.

INSTRUCTIONS

1. Create sets for Customer A and Customer B interests.
2. Calculate shared interests, all interests, A-only interests, and interests in exactly one set.
3. Create a list containing two customer dictionaries with nested contact dictionaries.
4. Retrieve the second customer's email and the first customer's first interest.
5. Explain why sets remove duplicates and why they do not preserve a useful business order.

REQUIRED SUBMISSION

- Four correct set-operation results.
- A valid list of nested customer records.
- Two correct nested lookups and a short explanation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Final submission checklist

- ☐ All five challenges are complete.
- ☐ The notebook runs from top to bottom without errors.
- ☐ Every output is clearly labelled.
- ☐ Important results include written explanations.
- ☐ Validation checks are included.
- ☐ Supplied raw data has not been overwritten.
- ☐ Sources and assistance are acknowledged.

Student declaration

I confirm that this submission is my work, that I have acknowledged sources and assistance, and that the displayed outputs were produced by the submitted code.

Signature: _____ Date: _____